ID Layers in GnuCash (Users and Programmers)

Overview

Generally speaking, you normally have two ID layers in a software, i.e. two layers to identify objects:

- the technical layer
- the business-logic layer

A user normally only deals with the business-logic layer, whereas a programmer would typically work primarily with the technical layer and provide services for the business-logic layer only indirectly, by translating between it and the technical layer.

In GnuCash, this system is used for all entities except one: the securities.

Security IDs in GnuCash

Figure 1 shows the two levels of security IDs in GnuCash.

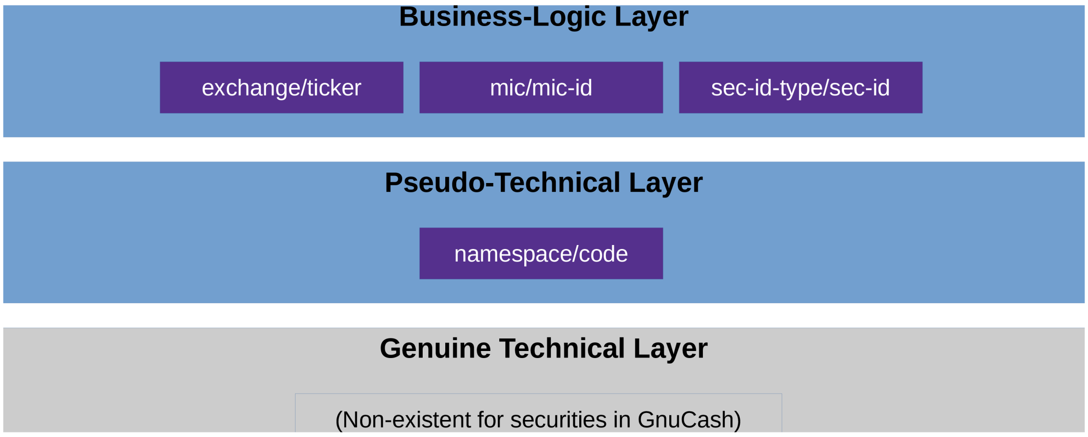

Figure 1: ID layers in GnuCash

In theory, the two layers mentioned in the previous section (i.e., the lowest and the highest one in figure 1) are clearly separated, i.e. technical IDs and business-logic ID do not have anything to do with each other. However, the GnuCash developers chose to make things a little more complicated: They intentionally blurred the line between the two layers, so that technical IDs are, in fact, pseudo-technical and quasi-business. More precisely:

- *Pseudo-technical level*: Securities have no technical IDs in GnuCash. Instead, they are technically selected with a (pseudo-)technical ID, constisting of a namespace and a code.
- *Business-logic level*: In the real world out there, you normally would identify a security by its public security code, which hopefully you have used when entering the security in the GnuCash file:
  - If you live in the US or Canada, then you would typically use the 1-to-4-char ticker (such as “T” or “MSFT”). But strictly speaking, this is not enough. It always has to

  be qualified with the according exchange (such as “NYSE_AMERICAN” (formerly known as “AMEX”) or “NASDAQ”). (The pros among you might use the CUSIP intead.)
- If you live outside of the US and Canada, then you would typically use something else. In the EU, where the current maintainer lives, people typically use the ISIN (international). In Germany, the nostalgic folks prefer the WKN. Similarly, in the rest of Europe: The SEDOL (UK) or the VALOR (Switzerland), etc. etc.

If you have a global portfolio, it makes sense to use the ISIN.[1]

So, if the developers had chosen to treat securities as any other entity in GnuCash, then a security's technical ID would look something like this: `14305dc80e034834b3f531696d81b493`. This string obviously has no meaning, i.e. it cannot be “interpreted” or “understood” nor is it meant to be.

Well, they chose another approach: In GnuCash, a (pseudo-)technical security ID looks something like these:

- `EURONEXT:SAP`
- `ISIN:DE000BASF111AP`

Just by glancing at these, you can see that they are essentially business-logic IDs maskerading as technical ones; they do mean something (i.e., they have semantics) and can thus very well be interpreted and understood.

It is important to understand this (non-)difference between technical and business-logic IDs in GnuCash before reading the other documents; they might otherwise confuse you.

You can also see that the two examples above are composed of two parts, which is no coincidence but rather the system that GnuCash imposes: `<namespace>:<code>`. Whereas the name space can, in theory, be freely chosen by the user, in practice it makes sense *not* to do so but instead to choose from a pre-defined set that `JGnuCashLib` provides. This then leads to the concept of providing IDs “indirectly”, i.e. composing them: providing the name space and the code separately.

[1] This is how thes current maintainer does it in his own portfolio, and it's been working well for decades now.